

Yuze Fu

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 Website |  LinkedIn |  Github

RESEARCH INTERESTS

Physics-based Animation, Geometry Processing, Numerical Methods for Partial Differential Equations

EDUCATION

- **University of Toronto** Sep 2023 - June 2027 (expected)
Honours Bachelor of Science in Mathematics and Computer Science
Toronto, Ontario, Canada
 - GPA: 3.99/4.00
 - Double Major in Mathematics and Computer Science
 - Math Courses: Abstract Algebra, Numerical Methods for PDEs, Number Theory, Graph Theory, Analysis
 - CS Courses: Machine Learning, Operating System, Database, Human-computer Interaction
- **High School Affiliated to Fudan University** Sep 2020 - June 2023
High School Diploma
Shanghai, China
 - Member of the school's competitive programming team for the Informatics Olympiad
 - Member of the school's football team, playing against other high schools in Shanghai

RESEARCH

- **Conservative Cell Splitting for Moving-Mesh Fluid Simulation** May 2026 - Present
Supervised by Eitan Grinspun and Jonathan Panuelos
 - Developing adaptive cell splitting for a moving-Voronoi fluid solver (incompressible and compressible), adding resolution where the flow needs it, implemented in C++ as Houdini plugins.
 - Showed the standard volume-based split introduces first-order density and pressure errors on the neighboring cells that re-tessellate around each split.
 - Designed a conservative remap that transfers mass, momentum, and energy by geometric cell overlap, removing those errors and keeping refinement exactly conservative on uniform flow.
 - Made the remap stable on moving, deforming meshes via exact-predicate geometry and a density-bounding normalization, validated with up to an order of magnitude lower error than the standard split.

EXPERIENCE

- **Dynamic Graphics Project Lab** May 2026 - Aug 2026
Research Intern
Toronto, Ontario, Canada
 - Worked on fluid simulation, funded by the DCS award.
- **Rotman School of Management** Jun 2025 - Aug 2025
Research Assistant
Toronto, Ontario, Canada
 - Worked on machine learning pipelines detecting misinformation in social media.

TEACHING

- **Teaching Assistant for CSCB09 (System Programming)** Winter 2026
- **Teaching Assistant for CSCB36 (Introduction to the Theory of Computation)** Fall 2025
- **Teaching Assistant for MATA22 (Linear Algebra I for Mathematical Science)** Winter 2025
- **Teaching Assistant for CSCA67 (Discrete Mathematics)** Fall 2024

HONORS AND AWARDS

- **Department of Computer Science (DCS) Award** 2026
University of Toronto \$12000
- **University of Toronto Scholar** 2024
University of Toronto \$1500
- **Dean's List** 2024, 2025, 2026
University of Toronto Scarborough
- **Second Prize in the National Olympiad in Informatics (NOIP2020)** 2020
China Computer Federation

PROJECTS

• **Classification of Finitely Generated Modules over PIDs**

Mar 2026 - Apr 2026

Abstract Algebra Essay

- Classified finitely generated modules over PIDs to unify the structure of abelian groups and operators.
- Formulated the bijection between $F[x]$ -modules and linear maps to prove operator structural identities.
- Constructed Rational and Jordan Canonical Forms using Invariant Factor and Elementary Divisor forms.
- Concluded with a review of the theoretical and computational trade-offs between different matrix canonical forms.

• **Numerical Analysis of Finite Difference Schemes for Hyperbolic PDEs**

Feb 2026 - Apr 2026

Numerical Methods for PDEs Projects

- Implemented 8 finite difference schemes using vectorized NumPy operations, optimizing performance for advection-based simulations.
- Evaluated the stability and order of convergence for standard schemes (e.g. Lax-Wendroff, Leapfrog) under varying CFL conditions.
- Developed SBP-SAT boundary treatments for high-order Runge-Kutta methods, ensuring stable and consistent convergence up to the 4th order.
- Analyzed theoretical vs. experimental trade-offs between ghost cell extrapolation and SBP-SAT operators for boundary stability.

• **NSI Forecasting: Machine-Learning-Driven Neighborhood Safety Index**

Nov 2025 - Dec 2025

Machine Learning Project

- Engineered a custom Neighborhood Safety Index (NSI) based on the Canadian Crime Severity Index to quantify urban safety levels.
- Evaluated Ridge, KNN, and FNN models to compare baseline linear accuracy against non-linear neural approximations.
- Developed an LSTM network achieving a leading R^2 of 0.8295 by modeling complex 12-month temporal sequences.
- Optimized performance through grid search tuning of architecture depth, activation functions, and k-neighbors.

COMMUNITY INVOLVEMENT

• **UTSC Accessibility Services**

Sep 2024 - Present

Volunteer Notetaker

- Produced comprehensive lecture and tutorial notes to support students with diverse accessibility needs.
- Courses: Probability (STAB52), Computation Theory (CSCB36), Group Theory (MATC01), Numerical Methods for PDEs (MATD10)

• **UTSC Summer School**

Jul 2025 - Aug 2025

Teaching Assistant

- Helped Professors to hold seminars about computer science for high school students.
- Answered students' questions and provided information about university lives.

• **The Association of Mathematical and Computer Science Student (AMACSS)**

Sep 2024 - Apr 2025

Course Representative

- Developed review materials and facilitated pre-exam seminars to improve peer academic performance.
- Courses: Discrete Math (CSCA67), Introduction to Computer Science (CSCA08)

• **UTSC Google Developer Group (GDG)**

Oct 2024 - Apr 2025

Academic Director

- Supported the execution of "Build with AI," an industry-focused event with 500+ registrations and live AI demos.
- Created an interactive emotion-matching game that successfully increased participant engagement.

SKILLS AND HOBBIES

- **Programming Languages:** C/C++, Python, Assembly, SQL, Shell Script, Java, R, LaTeX, Markdown
- **Frameworks and Libraries:** pandas, NumPy, SciPy, PyTorch, scikit-learn
- **Developer Tools:** Houdini, Docker, PostgreSQL, Git
- **Hobbies:** Soccer, Piano, Swimming, Traveling, Movie, RPG Game